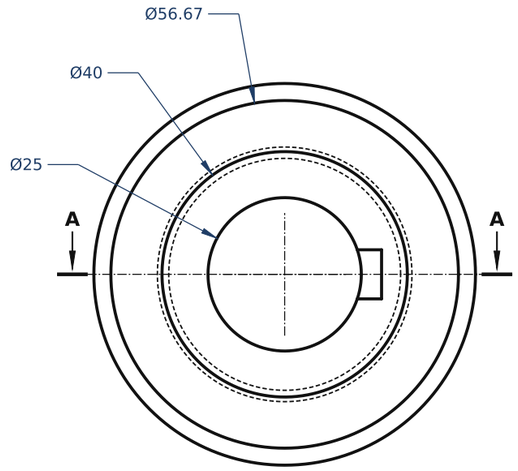
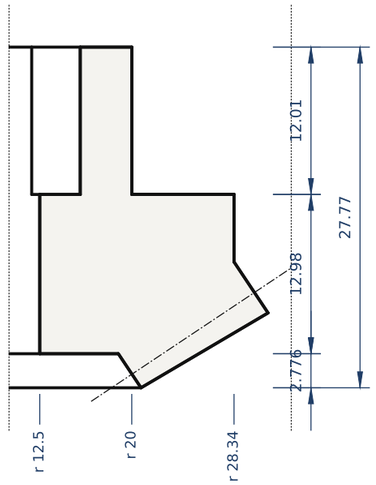
PLAN 1:1



SECTION A–A 2:1

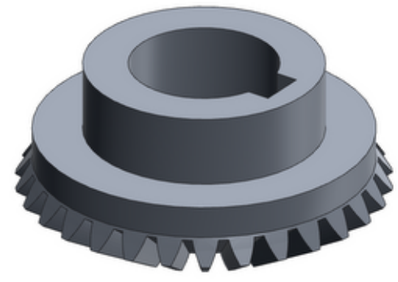
(its part from r 10 to 33)



REVISIONS

Rev	Date	Description
A	2026-10-11	First issue, draft for quotation

status	PROPOSED · the first cut, on the record's 940 mm stack-up: to re-cut on the layout (ledger §6, item 71)
qty per machine	1
used in	DRV-100
source	docs/drive-gear-belt.md §5; sim/drive_sizing_results.json gear
gear	straight bevel, 90° shaft angle; the teeth's proportions at the heel (h_a = m, h_f = 1.25 m)
teeth, module	z 30, m 2 at the heel; pressure angle 20°
pitch angle δ	56.31°; face 59.48°, root 52.34°
pitch Ø / tip Ø	60.00 / 62.22 at the heel
cone distance, face	R_e 36.06; b 12
mounting distance	28.00 ± 0.05, the cone's apex to the back face
mates with	DRV-103 pinion z 20, × 3
backlash	j_n 0.10-0.15, set at assembly (docs/drive-gear-belt.md §5)
accuracy	ISO 17485 grade 8 class (AGMA Q 8, as sized) [IR]
keyway	8 P9 × 3.3 in the hub only (DIN 6885 A 8 × 7)
mass each	0.268 kg



3-D view (axonometric, not to scale)

NOTES

1 The first cut's side gear (§5): z 30, m 2, b 12, 1.4305 (non-magnetic, as the shaft); the wheel of a 1.5 : 1 set, re-machined to this drawing, or cut from bar.
2 Keyed on the shaft's Ø25 journal in its own hub Ø40 × 12, not under the teeth: the small-end root is Ø38.2 (§5). Keyway 8 P9 × 3.3 in the hub only.
3 The mounting distance sets the mesh: shim the spacer under the hub (DRV-190) for the backlash.
4 The teeth as the 3-D view and the model draw them: the heel's involute profile on the back cone (Tredgold), the flanks straight to the apex, each tooth thinned 0.064 mm for the backlash (sim/unmodelled_parts.py); the section and the plan draw the blank, as a gear drawing does.
5 Status: the first cut of docs/drive-gear-belt.md on the record's 940 mm stack-up (z from the record's bottom end-bearing slot); the ledger asks for the drive to be re-cut on the layout (§6, item 71). PROPOSED.
6 General tolerances ISO 2768-mK; dimensions in mm; do not scale. Views by hidden-line removal of the model's solids.

DCCREG turbine · the drive: the reverser

Side gear, the rotor's, z 30 m 2, keyed		Drawing no.
Material: austenitic stainless 1.4305 (303), non-magnetic, as the shaft		**DCCREG-DRV-101**
Mass: 0.268 kg	Qty per machine: 1	Rev A Sheet 73 of 103 A3
Scale: 1:1, 2:1	Tolerances: ISO 2768-mK	Date: 2026-10-11
Drawn: generated by docs/make_drawing_set.py	Checked: — Approved: —	**PROPOSED · DRAFT**